UA LITTLE ROCK

MS Applicant (Fall 2026)

Maaz Bin Fazal

- CGPA: 3.31/4
- Duolingo English Test : 120/160 (IELTS 6.5)
- Research Projects / Publications: 4

Recognized By:

THE UNIVERSITY OF CALIFORNIA BERKELEY 1868 LET THERE BE LIGHT

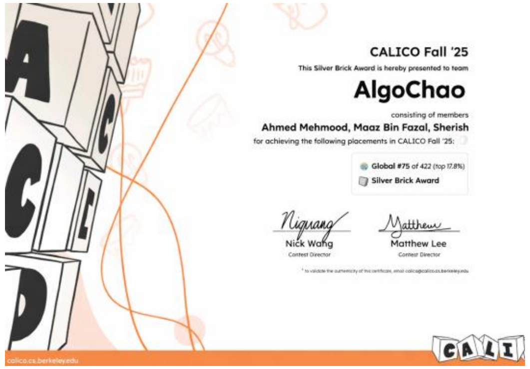
Stanford University

Meta

VE RI TAS HARVARD

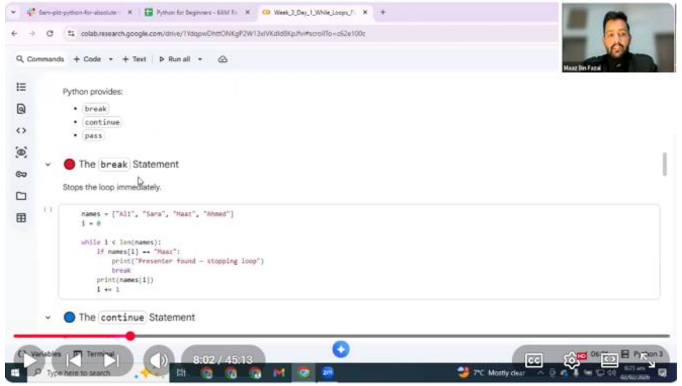
About me

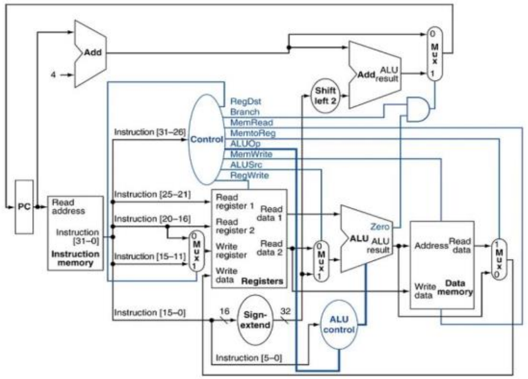
- Ranked Top 75 globally in UC Berkeley's Coding Challenge. [Link]
- Winner of Harvard CS50x Puzzle Day 2026 solved 10/10 puzzles. [Link]
- Qualified for Round 1 Meta Worldwide Algorithmic Programming Contest. [Link]
- Received Merit Based Scholarship. [Link]
- Competed in 5 International Hackathons.

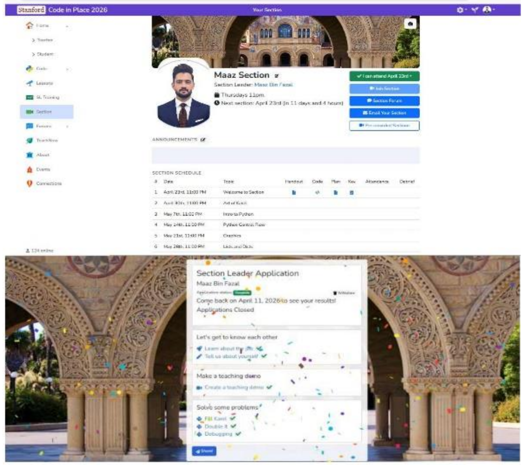


Teaching EXPERIENCE

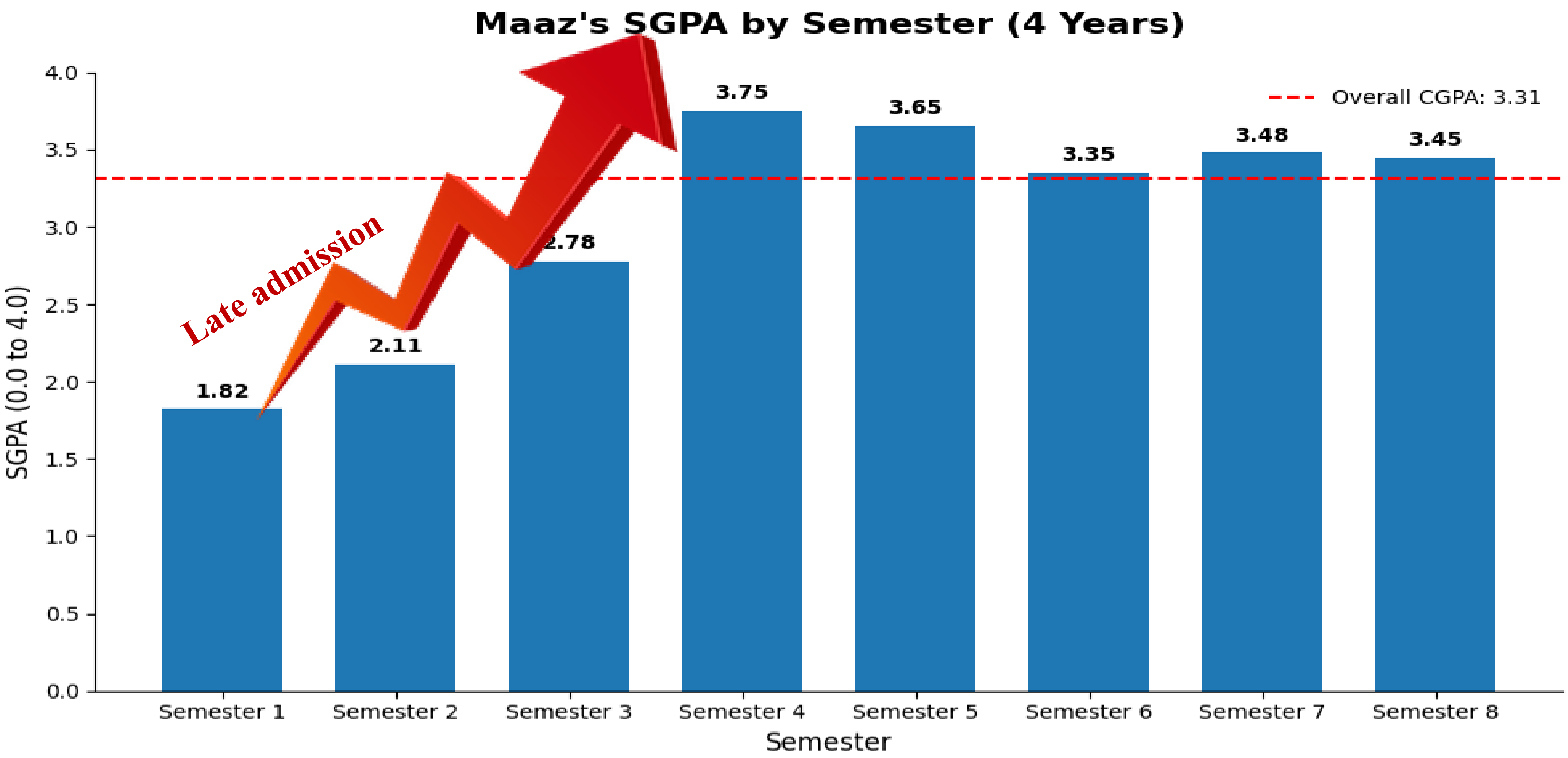
- Section Leader for CS106A (Python Course) in Stanford Code in Place 2026.
- Delivered 20 hours of Python live sessions to 30+ students. [YouTube Playlist]
- TA Computer Architecture: Taught Topics like pipelining, memory hierarchy, and cache design.
- Guide 200+ underprivileged students for STEM pathways.
- Contributed to 7 + Research articles.







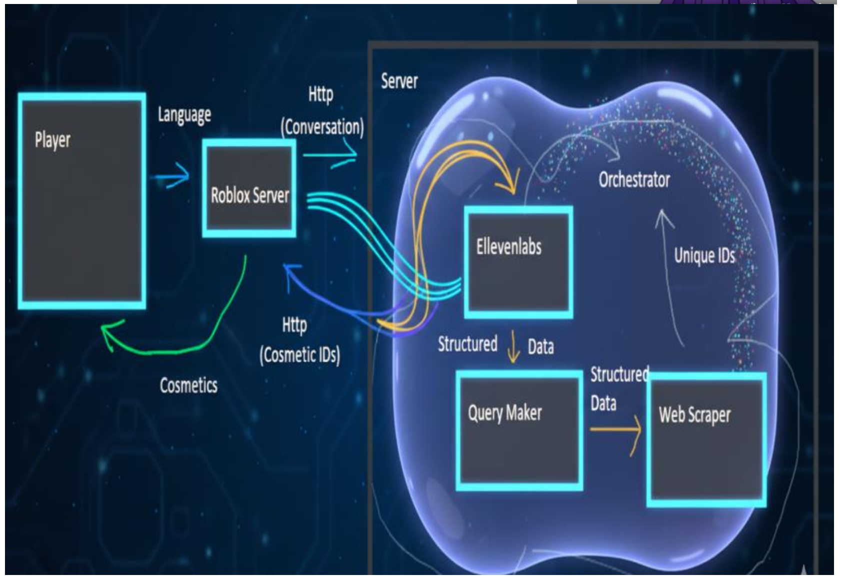
Academic Performance



INTERNATIONAL HACKATHONS

- Title: Robo Styler Agent – ETH Global AI Agent Hackathon
- Problems: Search is tedious and keyword bound.
- Solution: Integrate NLP to generate outfits.
- Result: Faster outfit automatic discovery.
- Technologies used: GPT 4, MultiAgent System,

Eleven Labs Open AI



FPGA Based Vending Machine

Objective

- ❖ Design FSM for vending machine control
- ❖ Implement using Verilog HDL on FPGA hardware
- ❖ Returns correct change automatically

Design Approach

- ❖ State based decision logic
- ❖ Sequential control using clocked design
- ❖ Simulate and verify functionality

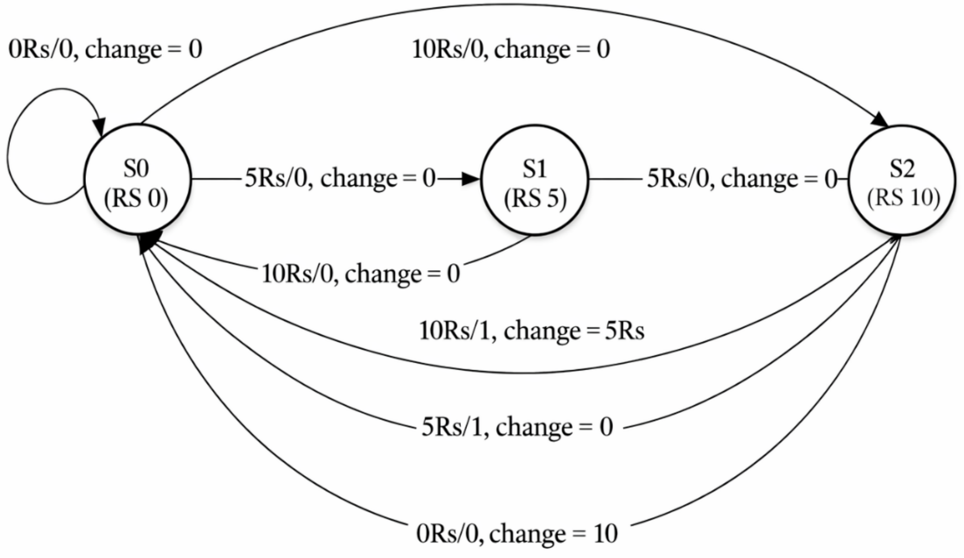


Figure 2: Vending Machine state diagram.

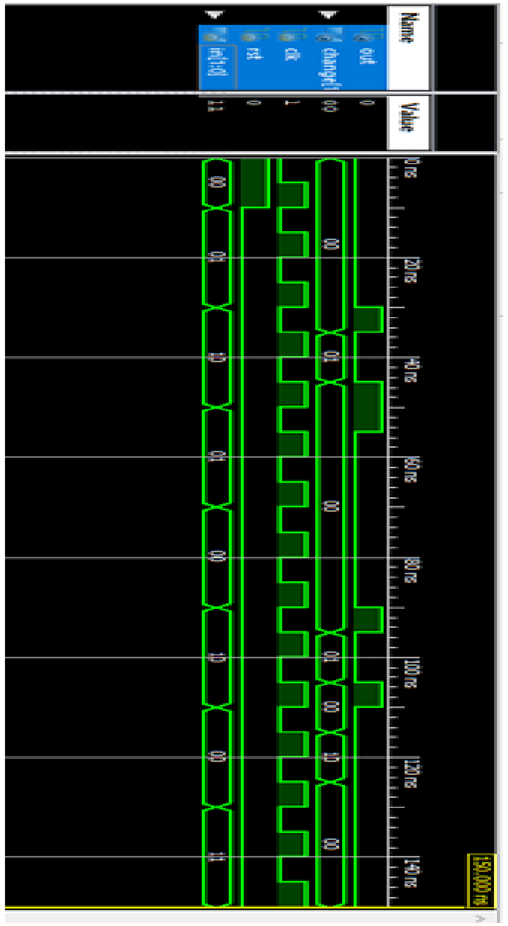
Simulation & Hardware Results

Results

- Correct FSM state transitions verified
- Accurate output & change return logic works correctly
- Stable real time FPGA execution

FPGA Implementation

- Demonstrates hardware level control using FSM on FPGA
- Implements synthesizable logic for real time systems
- Foundation for scalable embedded/edge FPGA applications



FPGA Result

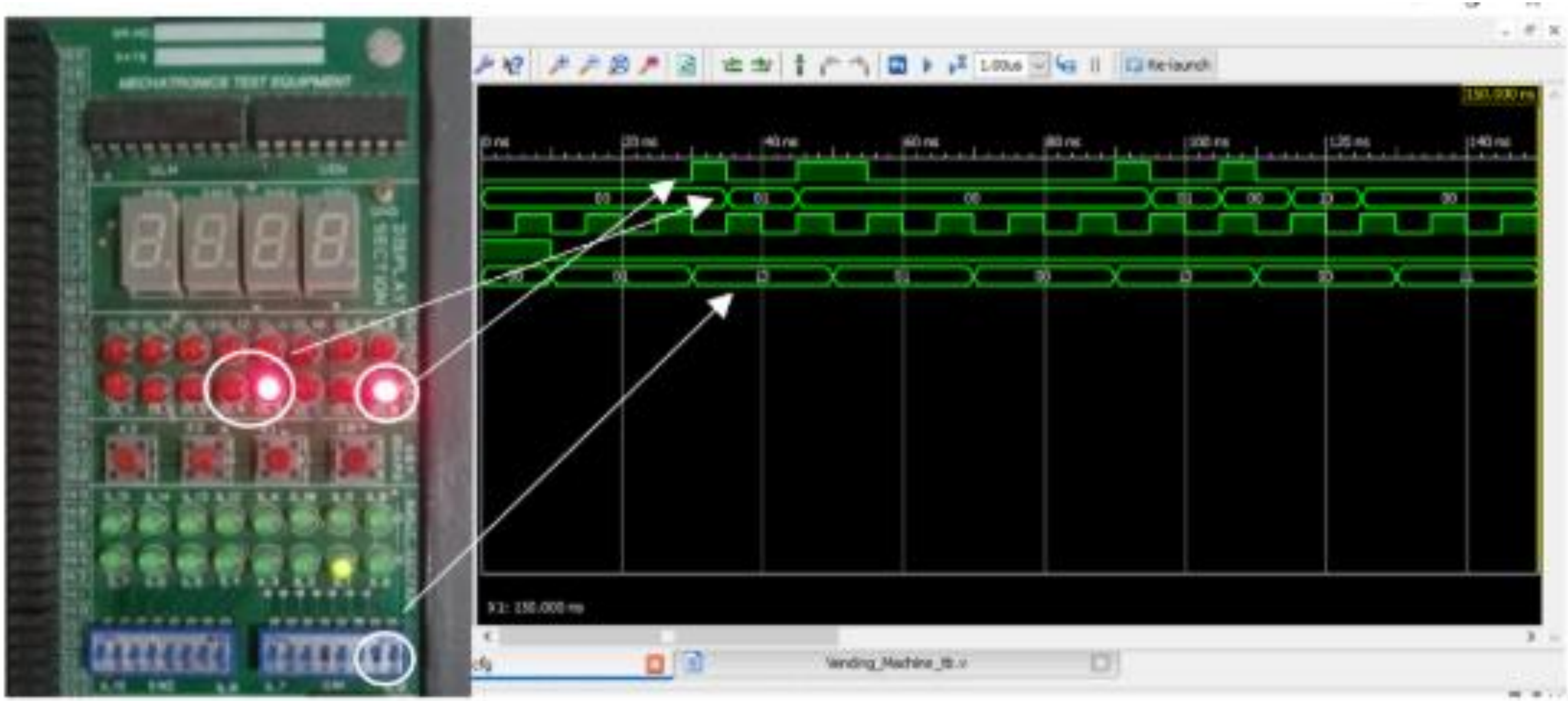


Figure 14: At input of Rs10 after adding two time Rs 5.

Senior/Final Year Project

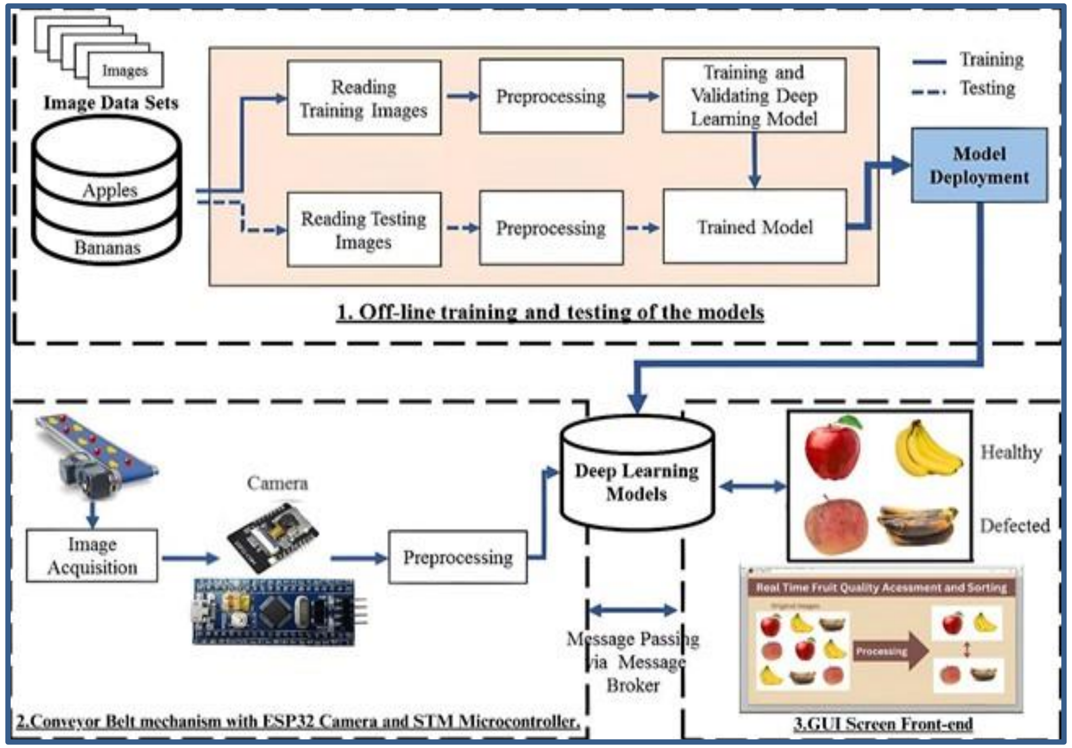
Title: **Low Cost Real Time Fruit Quality Assessment and Sorting Using Yolov8**

Problems:

- ❖ **Deep models unsuitable for microcontrollers**
- ❖ **Manual inspection slow & inconsistent**
- ❖ **High hardware cost in industry**

Result:

- ❖ **INT8 quantized lightweight deployment**
- ❖ **~200ms low-latency inference**
- ❖ **90%+ mAP achieved**

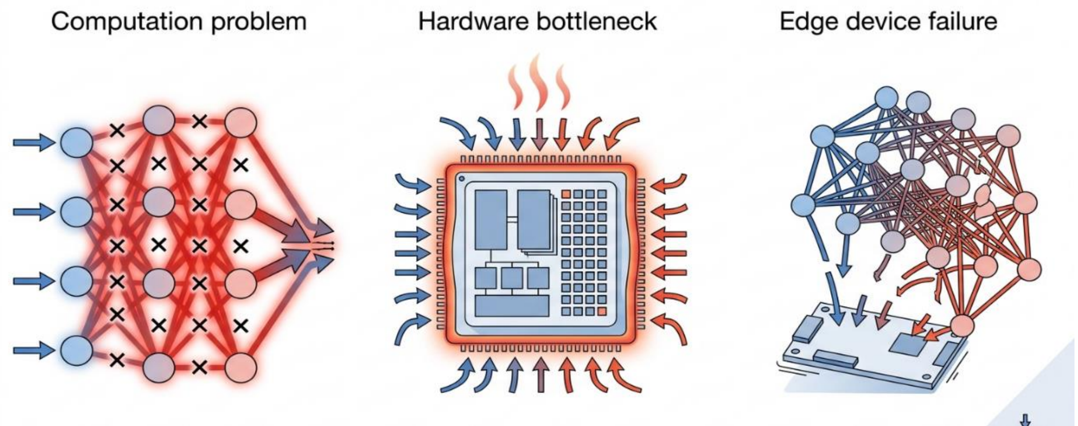


Research Interest

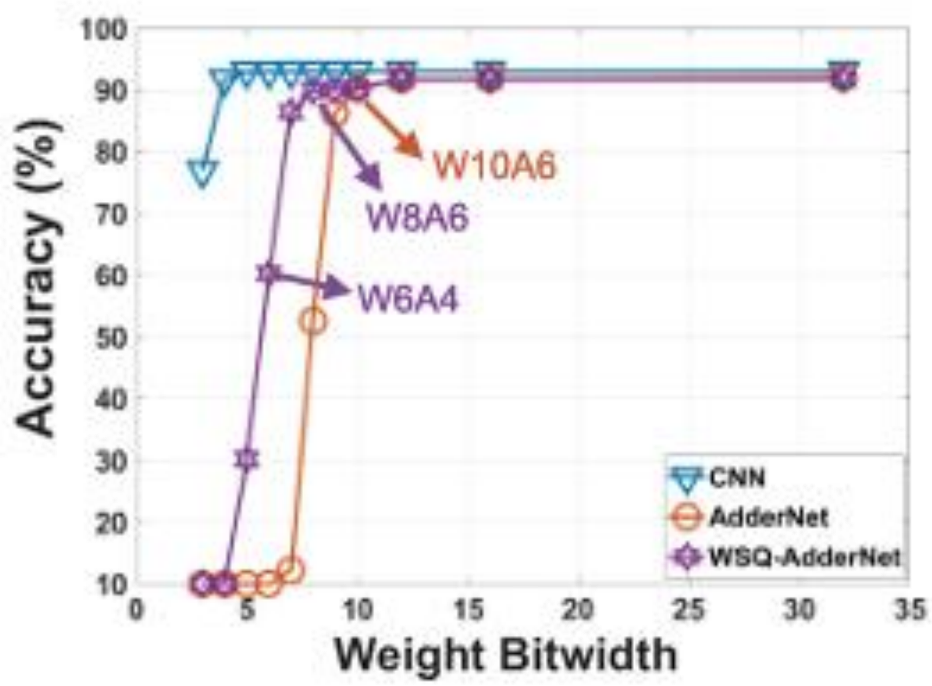
Title: **AdderNet 2.0: Optimal AdderNet Accelerator Designs With Activation-Oriented Quantization and Fused Bias Removal-Based Memory Optimization**

Motivation / Problems:

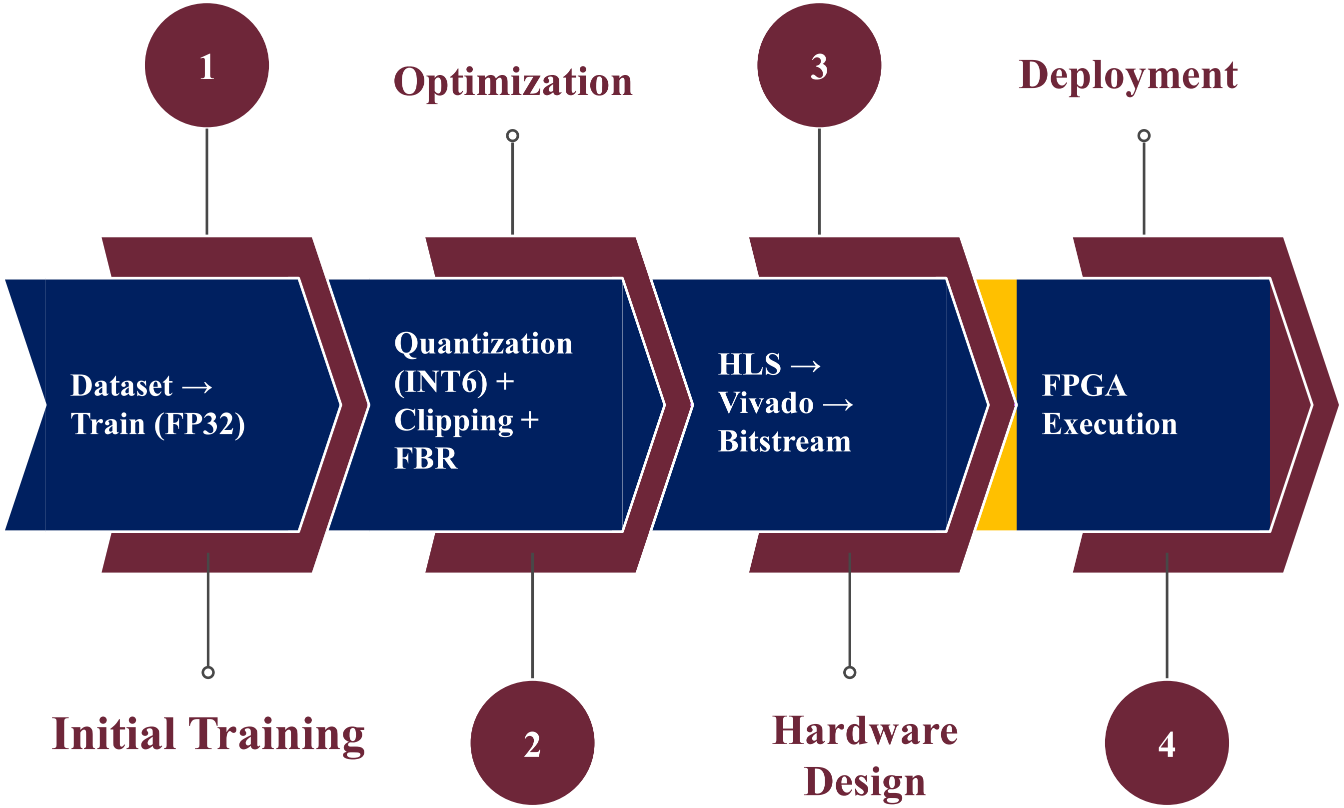
- ❖ **MAC heavy, power hungry computation**
- ❖ **High DSP and hardware cost**
- ❖ **Poor edge device efficiency**



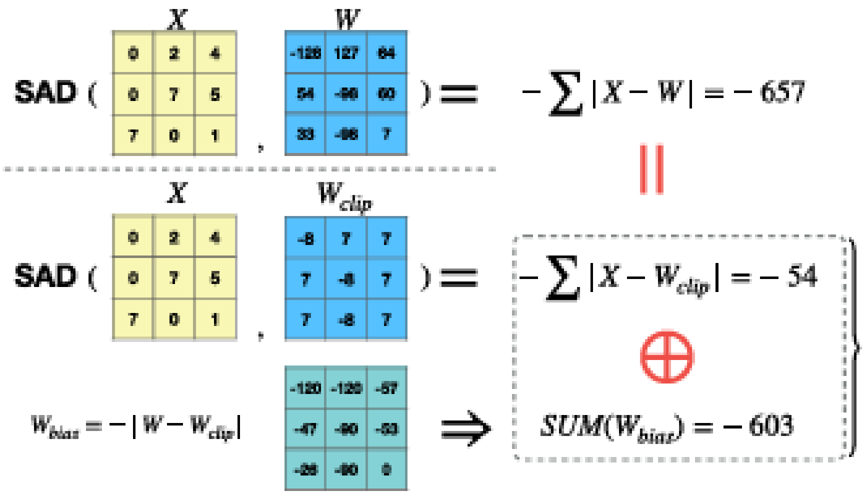
Challenge I: Model Quantization

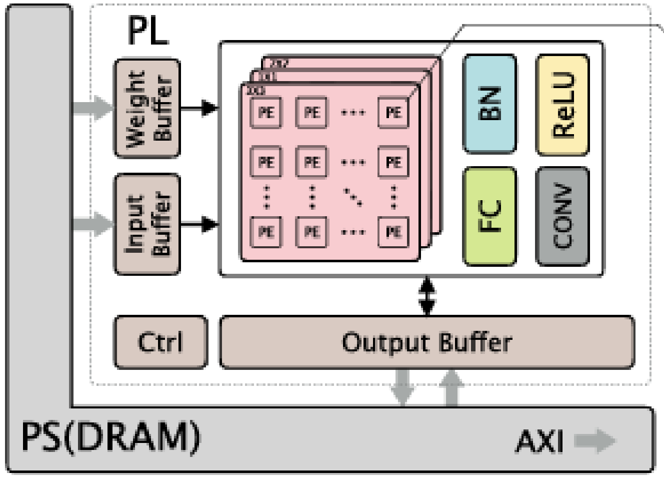


Workflow

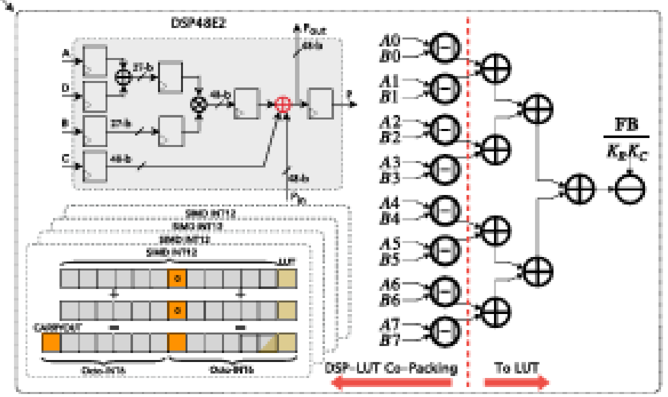

1
Initial Training
Dataset →
Train (FP32)
2
Optimization
Quantization
(INT6) +
Clipping +
FBR
3
Hardware
Design
HLS →
Vivado →
Bitstream
4
Deployment
FPGA
Execution

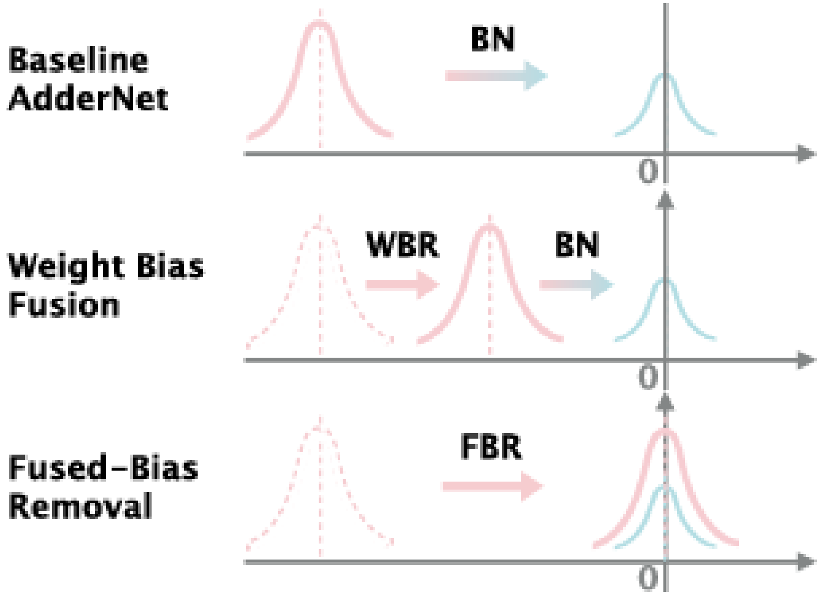
Proposed Architecture (Adder net 2.0)





FBR-SAD PE





How I can Contribute?

Limitations (FPGA focused)

- BN dependent FBR limits deployment in optimized inference pipelines
- Memory/dataflow inefficiency remains (bandwidth bottleneck not addressed)
- Hardware scalability not validated for larger/deeper networks

What value I can add ?

- FPGA/RTL experience (Verilog, FSM design) relevant to efficient accelerator development
- Edge AI deployment experience (INT8 quantization, real time constraints) relevant to FPGA accelerators
- System level perspective (algorithm to FPGA execution) for efficient ML to hardware mapping

LONG TERM GOAL

- ❖ **Begin my MS research journey**
- ❖ **Contribute to NSF funded AI projects**
- ❖ **Publish FPGA AI research in IEEE/ACM**
- ❖ **Advance scalable edge AI systems**
- ❖ **Pursue a PhD and academia**
- ❖ **Build technology that benefits humanity**

Thanks!

Looking forward to your thoughts and feedback

LET'S CONNECT in

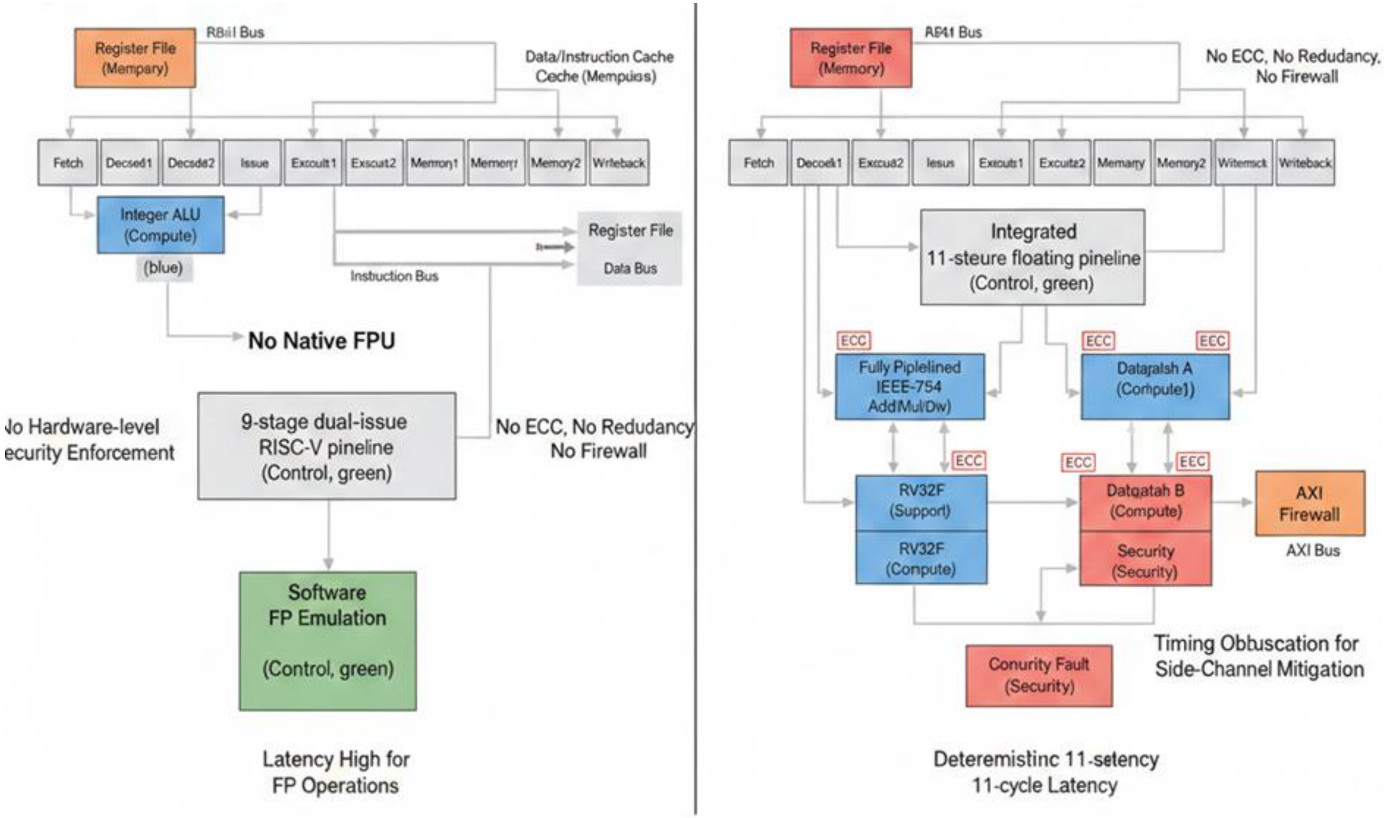
UA LITTLE ROCK

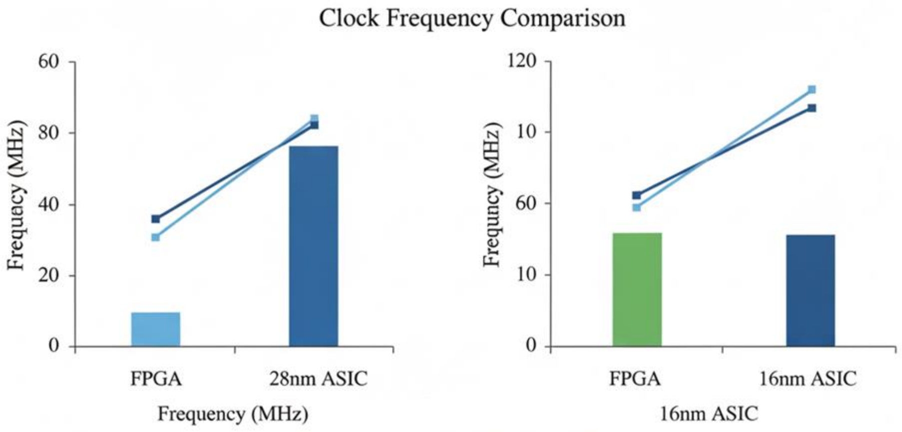
Current Work

Title: **Trust Enhanced RISC-V IEEE-754 FPU: ECC, Redundancy and Firewall Protected Execution for SweRV EH1**

Problems:

- No native SweRV FPU
- High latency software FP
- Security risks in hardware





Simulation Results